

Trajectory-Resolved Spectral Flow of Monitored Domain-Wall States

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¹ *Class-A fermionic adaptive-circuit project*
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We study a static spectral-flow diagnostic of pure Gaussian states prepared by the monitored class-A circuit. Each trajectory supplies an occupied projector, from which we construct a spectrally flattened one-particle parent and thread one transverse flux quantum. A previous-projector-overlap rule defines the continued occupied subspace. For $N_x \times N_y = 20 \times 24$ and $n_{\text{shell}} = 1$, the 64-interval continuation produces a sharply bimodal response: 71 of 100 soft-wall states and 36 of 100 hard-wall states transfer nearly one unit of charge, while the remaining states close at zero. The raw mean endpoints are $(+0.70740, -0.70716)$ for the soft wall and $(+0.35937, -0.35940)$ for the hard wall under counterclockwise and clockwise threading, respectively. The same endpoint projectors yield 64 and 30 unit events on a 128-interval grid. An independent 24×24 ensemble gives 91 soft-wall and 46 hard-wall events on the 64-interval grid. These results establish trajectory-resolved wall spectral flow, but they also show that the global overlap selector is sensitive to how a finite-width avoided crossing is resolved. The binary response is therefore a property of the specified continuation protocol, not by itself a grid-converged topological index or a real-time monitored pump.

I. THE QUESTION BEING ASKED

The monitored circuit prepares a pure Gaussian state rather than a ground state of a microscopic Hamiltonian. The question here is deliberately narrow: after one trajectory has prepared such a state, does its occupied one-particle subspace carry the same wall-to-wall spectral flow as a Chern insulator under flux insertion?

This question separates naturally into two parts. The first is the state prepared by the circuit. The second is the rule used to continue that state through flux. The charge observable is simple once this path has been fixed; the subtlety lies in defining the path. In particular, the calculation below does not evolve the monitored circuit while the flux changes. It constructs a static family of trajectory-dependent flattened parents and continues an occupied spectral subspace through that family.

The distinction matters. A nonzero endpoint response in this calculation is evidence for spectral flow of the prepared projector. It is not an unbiased Born-ensemble current, and it is not a claim that a particle was observed to cross the sample during monitored time evolution.

II. MONITORED ENDPOINT ENSEMBLE

For each wall construction and each sample $s = 0, \dots, 99$, the canonical CPU entry point `classA_U1FGTN.run_markov_circuit` generates an independent trajectory. The primary ensemble uses

$$(N_x, N_y) = (20, 24), \quad T = 2N_y = 48, \quad (1)$$

$$n_{\text{shell}} = 1, \quad (\alpha_1, \alpha_2) = (1, 30). \quad (2)$$

The initial state is a random pure Gaussian state at half filling. The overcomplete-Wannier (OW) measurements are applied in raster- y order with perfect correction, no postselection, and complex128 arithmetic. The two walls lie at $x = 5$ and $x = 15$. The soft construction retains the

full spatial mass profile, whereas the hard construction terminates OW support at the active slab.

One complete monitored trajectory is the independent sampling unit. The deterministic seed is derived from root seed 2026090304, the wall type, and the sample index. Soft- and hard-wall trajectories are independent rather than paired. Perfect feedback correction preserves the intended circuit rule, but it need not preserve the particle number of an individual Born trajectory. We therefore retain the final occupied rank r_ξ of every trajectory ξ instead of projecting all samples back to a common rank.

III. FROM AN OCCUPIED FRAME TO A THREADED PARENT

Let

$$F_\xi \in \mathbb{C}^{2N_x N_y \times r_\xi}, \quad F_\xi^\dagger F_\xi = \mathbf{1}_{r_\xi}, \quad (3)$$

be an occupied frame for trajectory ξ . Its columns are a basis, not additional physical degrees of freedom. The basis-independent object is the one-particle correlation projector

$$G_\xi = F_\xi F_\xi^\dagger, \quad G_\xi^2 = G_\xi = G_\xi^\dagger. \quad (4)$$

The implementation historically stores this matrix under the name $\mathcal{C}$; we use G here to retain the notation of the main technical work.

We associate to this projector the flattened one-particle parent

$$h_\xi(0) = \mathbf{1}_{2N_x N_y} - 2G_\xi. \quad (5)$$

The occupied and empty subspaces of G_ξ are then the -1 and $+1$ eigenspaces of $h_\xi(0)$. Equation (5) introduces no new state: $G_\xi = (\mathbf{1} - h_\xi)/2$. Its purpose is to turn occupation into spectral ordering. Once this is done, twisting the parent supplies both the occupied and empty

instantaneous eigenspaces needed to define a spectral continuation. The parent is trajectory specific and should not be confused with the microscopic controller Hamiltonian used during preparation.

The periodic direction is y . For one-particle labels $a = (x, y, \mu)$ and $b = (x', y', \mu')$, let $d_y(a, b)$ denote the minimum-image displacement on the N_y -site ring. In the uniform gauge we define

$$\tilde{h}_{\xi,ab}(\phi) = h_{\xi,ab}(0) \exp \left[\frac{i\phi}{N_y} d_y(a, b) \right], \quad (6)$$

$$h_{\xi}(\phi) = \frac{1}{2} \left[\tilde{h}_{\xi}(\phi) + \tilde{h}_{\xi}^{\dagger}(\phi) \right]. \quad (7)$$

The second line removes only floating-point anti-Hermiticity. It does not modify the intended flux family.

Clockwise (CW) and counterclockwise (CCW) paths use

$$\phi_j^{(\sigma)} = -\sigma\epsilon + \sigma \frac{2\pi j}{M}, \quad \epsilon = 10^{-7}, \quad \sigma = \pm 1, \quad (8)$$

with $j = 0, \dots, M$. The signed offset chooses a one-sided representative at the seam where an exact crossing would otherwise make the initial label ambiguous. It is not a physical flux ramp and cannot manufacture an integer response.

IV. PREVIOUS-PROJECTOR CONTINUATION

At flux point j , diagonalize

$$h_{\xi}(\phi_j) u_{a,j} = E_{a,j} u_{a,j}. \quad (9)$$

Suppose P_{j-1} is the rank- r_{ξ} projector selected at the preceding point. The weight

$$\omega_{a,j} = \langle u_{a,j} | P_{j-1} | u_{a,j} \rangle = \| F_{j-1}^{\dagger} u_{a,j} \|^2 \quad (10)$$

measures how much of the new eigenvector already lies in the previously occupied subspace. We select the r_{ξ} eigenvectors with the largest $\omega_{a,j}$ and define

$$P_j = \sum_{a \in \mathcal{I}_j} |u_{a,j}\rangle \langle u_{a,j}|. \quad (11)$$

Instantaneous energy enters only as a deterministic tie breaker.

This prescription can be stated without choosing a frame. Among projectors formed from r_{ξ} eigenvectors of $h_{\xi}(\phi_j)$, Eq. (11) maximizes

$$\text{tr}(P_{j-1} P_j) = \sum_{a \in \mathcal{I}_j} \omega_{a,j}. \quad (12)$$

The computed path is therefore a path of projectors. A frame is retained only as a numerically efficient representation of that path; arbitrary unitary rotations among its occupied columns do not change the result.

As a control, we also form the instantaneous projector P_j^{inst} from the r_{ξ} lowest eigenvalues independently at every flux. It closes after one flux quantum. The distinction between P_j and P_j^{inst} is thus not the final Hamiltonian—the two paths end at gauge-equivalent spectral problems—but the occupation labels carried through the crossing.

V. SUBSYSTEM CHARGE

Let R_L and R_R project onto the left and right half cylinders, respectively. For $N_x = 20$,

$$R_L = \sum_{x=0}^9 \sum_{y,\mu} |x, y, \mu\rangle \langle x, y, \mu|, \quad (13)$$

$$R_R = \sum_{x=10}^{19} \sum_{y,\mu} |x, y, \mu\rangle \langle x, y, \mu|. \quad (14)$$

The continued regional charges and their changes from the starting projector are

$$N_A(\phi_j) = \text{tr}(R_A P_j), \quad (15)$$

$$\Delta N_A(\phi_j) = N_A(\phi_j) - \text{tr}(R_A P_0), \quad A = L, R. \quad (16)$$

The transverse transfer is

$$q_x(\phi_j) = \frac{\Delta N_R(\phi_j) - \Delta N_L(\phi_j)}{2}. \quad (17)$$

Because the rank is fixed and $R_L + R_R = \mathbf{1}$, $\Delta N_L + \Delta N_R = 0$ up to roundoff. Hence $q_x = \Delta N_R = -\Delta N_L$. An endpoint $q_x \simeq +1$ is simply one unit removed from the left half and added to the right half in the continued projector. The reverse process gives $q_x \simeq -1$.

We report raw CW and CCW values separately. The paired combination

$$q_x^{\text{odd}} = \frac{q_x^{\text{CCW}} - q_x^{\text{CW}}}{2} \quad (18)$$

is used only as a compact measure of direction reversal and to reproduce the original event ledger. It is not substituted for the sample-resolved raw distribution.

VI. THE 64-INTERVAL RESULT

All 200 monitored endpoints and all 400 directional continuations completed for the 20×24 , $n_{\text{shell}} = 1$ ensemble. An independent readback audit verified every result/completion pair, comprising 1.331 GiB of output. The raw endpoint means and sample standard deviations are

$$\overline{q_x^{\text{CCW}}}_{\text{soft}} = +0.70740, \quad s_{\text{soft}}^{\text{CCW}} = 0.45442, \quad (19)$$

$$\overline{q_x^{\text{CW}}}_{\text{soft}} = -0.70716, \quad s_{\text{soft}}^{\text{CW}} = 0.45428, \quad (20)$$

$$\overline{q_x^{\text{CCW}}}_{\text{hard}} = +0.35937, \quad s_{\text{hard}}^{\text{CCW}} = 0.48157, \quad (21)$$

$$\overline{q_x^{\text{CW}}}_{\text{hard}} = -0.35940, \quad s_{\text{hard}}^{\text{CW}} = 0.48161. \quad (22)$$

The reversal is nearly exact: the largest samplewise direction-even endpoint is 6.31×10^{-3} for the soft wall and 1.68×10^{-3} for the hard wall.

The ensemble means hide the structure of the result. On the $M = 64$ grid, 71 soft-wall trajectories have $|q_x^{\text{odd}}| > 0.5$. Their conditional mean is 0.99617 and their minimum is 0.95733; every other soft-wall endpoint has magnitude below 1.60×10^{-8} . The hard-wall ensemble contains 36 such events, with conditional mean 0.99828 and minimum 0.98828; every remaining hard-wall endpoint again closes below 1.60×10^{-8} . The response is therefore a mixture of two trajectory-resolved sectors, not a fractional charge attached to a typical state.

The numerical residuals are far below either sector. Across all 400 paths, the largest violation of $\Delta N_L + \Delta N_R = 0$ is 2.84×10^{-13} , the largest instantaneous endpoint is 1.59×10^{-8} , and the largest large-gauge parent mismatch is 7.11×10^{-16} . Final particle-number rank has negligible linear association with the response (Pearson -0.073 for the soft wall and 0.020 for the hard wall). The zero/unit split is therefore not produced by charge nonconservation or by the raw variation of final rank.

VII. WHAT CONTROLS THE BINARY SECTOR

Let $E_{1,j} \leq \dots \leq E_{2N_x N_y, j}$ be the eigenvalues of the threaded parent and let r_ξ be the occupied rank. The spectral pair at the occupation boundary has gap

$$\delta_{\text{edge}}(\phi_j) = E_{r_\xi+1,j} - E_{r_\xi,j}. \quad (23)$$

For every 20×24 trajectory, the minimum occurs at $|\phi| \simeq \pi$. Unit-transfer trajectories have systematically smaller minimum gaps. The soft-wall medians are 0.00611 in the unit sector and 0.02652 in the closing sector; the hard-wall medians are 0.00673 and 0.02624. Using small gap as a descriptive classifier gives areas under the receiver-operating curve of 0.898 and 0.911, respectively. This is a strong association, but not yet a proof of a topological dichotomy.

The latter distinction is resolved by repeating the continuation on the *same* endpoint projectors with $M = 128$. Seven soft-wall samples and six hard-wall samples move from the unit sector at $M = 64$ to the closing sector at $M = 128$; no sample moves in the opposite direction. The event counts become 64/100 and 30/100. The corresponding raw endpoint means are $(+0.63777, -0.63750)$ for the soft wall and $(+0.29951, -0.29953)$ for the hard wall.

This is not ordinary quadrature error in Eq. (17). The finite grid is part of the continuation rule. Near an avoided crossing, the eigenvectors rotate over a flux scale set by the inter-wall hybridization. A step larger than that scale follows the same wall character across the exchange of energy ordering; a sufficiently fine step follows the smooth finite-size energy eigenprojector, which closes. The $M = 64$ binary distribution is therefore an exact result for the stated overlap protocol, but it is not invariant under refinement of that protocol.

VIII. FIRST TRANSVERSE-WIDTH COMPARISON

An independent $N_x \times N_y = 24 \times 24$ ensemble repeats the $M = 64$ protocol with $S = 100$ trajectories per wall, $n_{\text{shell}} = 1$, $T = 48$, and walls at $x = 6$ and $x = 18$. Table I collects the directly comparable raw endpoint means and unit-event counts.

At fixed $M = 64$, increasing the wall separation from 10 to 12 sites raises the soft-wall event fraction by 0.20 and the hard-wall fraction by 0.10. The closing-sector median gap rises from 0.02652 to 0.03367 in the soft ensemble and from 0.02624 to 0.04019 in the hard ensemble. The unit-sector medians at 24×24 are 0.00740 for both walls, while the closing-sector medians are 0.03367 and 0.04019.

The width comparison is suggestive but not yet a scaling law. The two sizes use independent monitored ensembles, only two values of N_x are complete, and the global overlap rule itself changes under grid refinement. The soft increase is large; the hard increase is smaller and statistically compatible with a broad range of width effects. No thermodynamic limit should be inferred from these two points.

IX. INTERPRETATION

The calculation has isolated a precise object. A monitored trajectory produces G_ξ . The flattened parent $h_\xi = \mathbf{1} - 2G_\xi$ promotes its occupied/empty decomposition to a spectral problem. The overlap selector then carries an occupation label through the threaded spectrum, and q_x reads the resulting left-right redistribution. Once the path is specified, there is no ambiguity in the charge measurement.

What is not fixed by q_x is the branch through the finite-width wall crossing. The exact translation-invariant flattened-Hamiltonian benchmark retains a clean wall crossing and gives $|q_x| = 0.981\text{--}1.000$ under refined continuation. A trajectory-derived parent breaks transverse translation symmetry and generically opens a sample-dependent avoided gap. The global $M = 64$ overlap rule often crosses this gap diabatically, producing one unit of wall transfer. Refining the grid resolves some of the same gaps and instead follows the smooth finite-size projector back to itself.

This exposes a noncommuting order of limits. The inter-wall gap falls with wall separation, whereas the edge-to-bulk separation remains finite. The spectral-flow limit is adiabatic relative to the bulk but diabatic relative to the exponentially small inter-wall splitting. Full finite-size adiabatic continuation takes the opposite limit and closes. A robust continuation must therefore identify the isolated two-wall subspace and preserve wall character inside it, rather than obtain the desired branch accidentally from a coarse global grid. That wall-diabatized calculation is documented separately and should be regarded as the controlled successor to the present protocol.

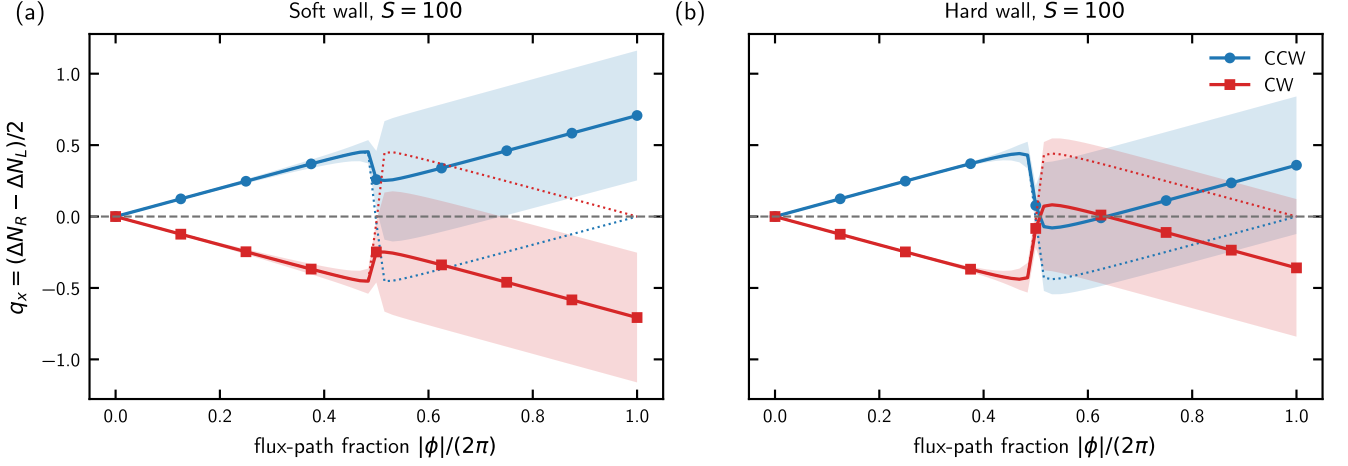


FIG. 1. Static projector response at $N_x \times N_y = 20 \times 24$, $n_{\text{shell}} = 1$, and $M = 64$. Each wall uses $S = 100$ independent monitored trajectories prepared from random pure half-filled states for $T = 48$ raster- y cycles. Curves are trajectory means and bands are one sample standard deviation. Solid curves show the previous-projector continuation; dotted curves show independently refilled instantaneous projectors. CW and CCW reverse the continued response, whereas the instantaneous control closes after one flux quantum.

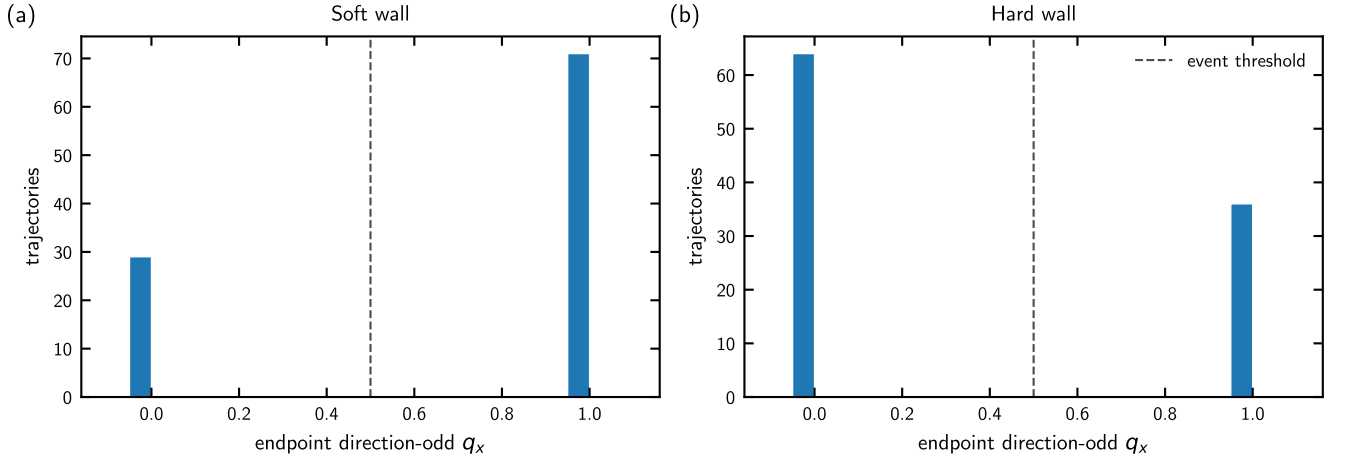


FIG. 2. Sample-resolved endpoint distribution for the same $S = 100$ ensembles. The plotted direction-odd coordinate only superposes the paired CW and CCW reversals; the archived table retains both raw endpoints. The dashed line at 0.5 defines the descriptive event count. It is not an acceptance threshold for topology or quantization.

The completed result supports three statements and no stronger ones. First, the prepared trajectory projectors retain wall-localized spectral structure. Second, the specified $M = 64$ overlap continuation produces sharply quantized samplewise transfer with nearly exact CW–CCW reversal. Third, the fraction of such events depends on the avoided-crossing scale, the flux mesh, and probably the transverse wall separation. It does not establish a quantized response of the trajectory-averaged state, a quantized monitored-time pump, or a universal finite-width event probability.

X. REPRODUCIBILITY

The 20×24 , $M = 64$ campaign is archived under `results/N20x24_state_projector_pump_s100_v1`; its locked configuration SHA-256 begins `aafda9bcd52995b48`. The exact same endpoints at $M = 128$ are under `results/N20x24_state_projector_pump_grid128_s100_v1`. The independent 24×24 , $M = 64$ ensemble is under `results/N24x24_state_projector_pump_s100_v1`. Each trajectory and each directional continuation is stored as an atomic NPZ/result pair with a checksum-bearing completion JSON. The CSV tables retain wall

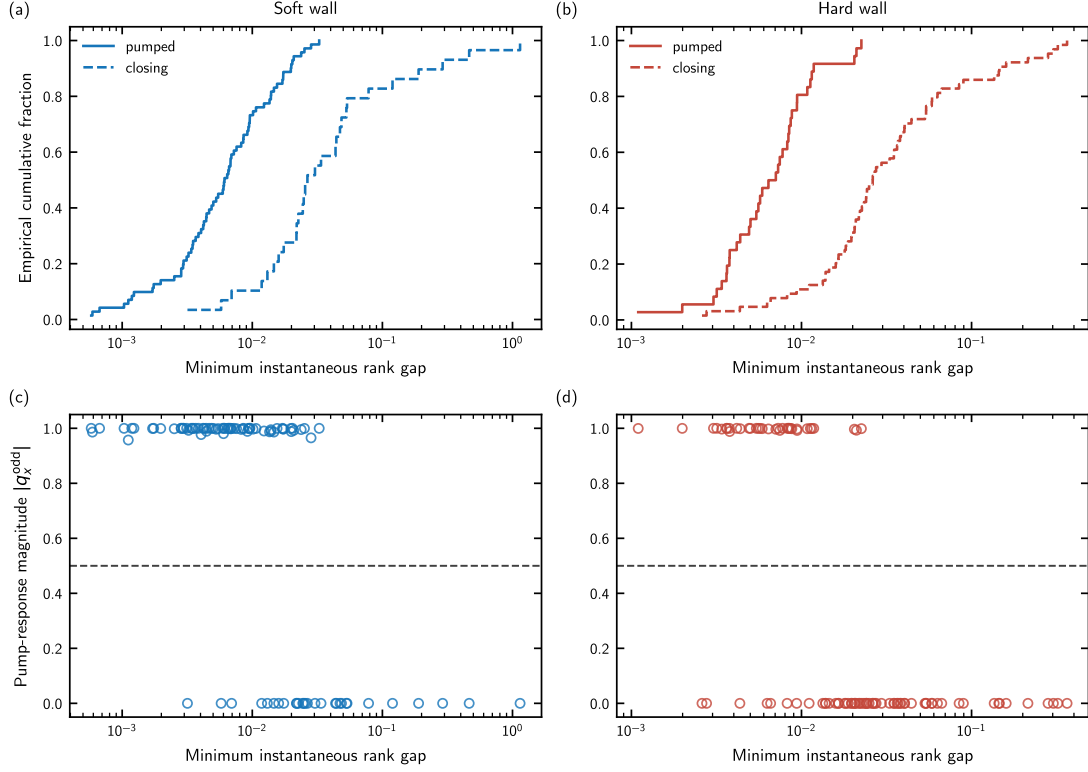


FIG. 3. Relation between the $M = 64$ endpoint sector and the minimum occupied–empty gap near $|\phi| = \pi$ for $S = 100$ independent trajectories per wall. The unit-transfer sector is concentrated at smaller gaps. This diagnostic cannot by itself distinguish an intrinsic crossing from a finite-grid branch-selection effect.

TABLE I. Completed overlap-continuation ensembles. Each row contains $S = 100$ independent monitored trajectories per wall. The event column uses the historical descriptive criterion $|q_x^{\text{odd}}| > 0.5$; the primary observables are the separate raw CW and CCW endpoints.

$N_x \times N_y$	M	$\overline{q_x^{\text{CCW}}}$ soft	$\overline{q_x^{\text{CW}}}$ soft	soft events	$\overline{q_x^{\text{CCW}}}$ hard	$\overline{q_x^{\text{CW}}}$ hard	hard events
20×24	64	+0.70740	−0.70716	71/100	+0.35937	−0.35940	36/100
20×24	128	+0.63777	−0.63750	64/100	+0.29951	−0.29953	30/100
24×24	64	+0.90749	−0.90726	91/100	+0.45896	−0.45903	46/100

type, sample identity, final rank, and separate raw CW and CCW endpoints. The crossing-gap analysis is

stored below the primary 20×24 analysis directory in `crossing_diagnostics_v1`.